

ZELTWEG, Austria

WMO: 111650

Lat: 47.200N

Lon: 14.750E

Elev: 682

StdP: 93.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.5	(c) -13.0	(d) -17.3	(e) 0.9	(f) -14.6	(g) -15.0	(h) 1.1	(i) -12.4	(j) 10.0	(k) 3.5	(l) 8.8	(m) 2.9	(n) 1.0	(o) 260	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	29.8	19.0	27.9	18.4	26.0	17.8	20.1	27.4	19.1	26.1	18.3	24.6	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	13.5	22.9	16.7	13.0	21.9	16.0	12.3	20.9	60.5	27.4	57.4	26.2	54.5	24.7	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 6.8	(c) 5.7	(e) -18.6	(f) 32.4	(g) 3.2	(h) 1.8	(i) -20.9	(j) 33.8	(k) -22.8	(l) 34.8	(m) -24.6	(n) 35.9	(o) -26.9	(p) 37.2		
(a) 8.2	(b) 6.8	(c) 5.7	(e) -18.7	(f) 21.7	(g) 3.0	(h) 1.3	(i) -20.9	(j) 22.6	(k) -22.7	(l) 23.3	(m) -24.4	(n) 24.1	(o) -26.6	(p) 25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-3.3	-0.4	3.8	8.2	13.1	16.8	18.4	17.9	13.5	8.7	3.3	-2.1
	DBStd	8.43	4.55	4.54	4.10	3.62	3.27	3.49	3.03	3.01	3.26	3.77	4.17	4.19
	HDD10.0	1637	414	291	192	74	9	1	0	1	8	68	202	377
	HDD18.3	3798	672	524	449	303	165	70	37	45	148	299	450	635
	CDD10.0	978	0	0	1	22	104	204	261	244	112	28	2	0
	CDD18.3	96	0	0	0	0	2	23	39	30	2	0	0	0
	CDH23.3	1060	0	0	0	2	58	264	383	322	31	1	0	0
	CDH26.7	271	0	0	0	0	8	65	103	92	2	0	0	0
Wind	WSAvg	1.9	1.6	1.8	2.4	2.3	2.3	2.2	2.1	1.9	1.7	1.6	1.6	1.4
Precipitation	PrecAvg	942	35	37	49	57	91	124	144	129	94	78	62	43
	PrecMax	1130	66	94	119	105	137	172	314	212	127	141	130	96
	PrecMin	714	7	7	12	16	43	69	25	67	51	10	0	4
	PrecStd	96	18	23	22	19	28	31	58	40	24	31	29	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	14.1	18.2	23.2	27.7	31.0	32.1	32.0	26.2	21.9	16.8	10.4
		MCWB	4.8	7.8	10.0	13.3	16.8	19.1	19.7	19.9	17.8	15.2	11.1	6.8
	2%	DB	6.5	10.8	16.0	20.8	24.8	28.7	29.6	29.2	24.0	19.2	13.8	7.2
		MCWB	3.7	5.7	9.1	11.9	15.7	18.6	19.1	19.2	16.7	13.8	9.7	4.4
	5%	DB	4.5	8.1	13.7	18.3	22.8	26.5	27.6	27.0	21.9	17.0	11.2	4.9
		MCWB	2.3	4.2	7.9	10.9	14.7	17.8	18.4	18.6	15.7	12.5	8.5	2.9
	10%	DB	2.8	6.0	11.2	16.1	20.4	24.2	25.5	24.9	19.8	14.9	9.3	3.1
		MCWB	1.0	2.8	6.4	10.0	13.7	16.8	17.7	17.7	14.8	11.6	7.3	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.7	10.9	14.2	17.9	20.7	21.4	21.2	18.5	15.8	11.7	7.5
		MCDB	8.1	13.3	16.9	21.8	25.4	28.3	28.8	28.8	24.8	20.8	15.3	10.2
	2%	WB	3.9	5.8	9.3	12.6	16.4	19.3	20.1	19.9	17.2	14.3	10.3	4.9
		MCDB	5.9	9.8	15.2	19.0	23.3	26.5	27.4	27.1	22.7	18.2	13.0	6.7
	5%	WB	2.4	4.5	8.1	11.4	15.2	18.3	19.1	19.0	16.2	13.0	9.0	3.2
		MCDB	4.3	7.9	13.1	17.2	21.4	25.0	26.0	25.8	20.8	16.3	11.3	4.5
	10%	WB	1.1	3.0	6.7	10.4	14.1	17.2	18.2	18.1	15.2	11.9	7.6	1.9
		MCDB	2.3	5.5	10.6	15.4	19.5	23.1	24.3	24.1	18.9	14.5	9.2	2.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	10.2	11.6	11.9	11.9	11.9	11.9	11.7	10.9	10.5	8.1	7.6
		MCDBR	9.9	13.3	16.4	16.8	16.4	15.9	15.6	16.0	14.3	13.8	11.1	9.9
		MCWBR	7.2	8.7	9.8	9.2	8.1	7.1	6.7	7.2	7.7	8.5	7.7	7.3
	5% WB	MCDBR	9.0	12.5	15.3	14.8	14.9	14.7	14.0	14.5	12.8	12.7	10.2	8.7
		MCWBR	6.7	8.4	9.4	8.2	7.6	6.7	6.4	6.9	7.0	8.1	7.2	6.6
Clear-Sky Solar Irradiance	taub	0.282	0.311	0.353	0.398	0.405	0.412	0.413	0.409	0.378	0.349	0.317	0.283	
	taud	2.418	2.358	2.312	2.249	2.276	2.295	2.292	2.321	2.382	2.441	2.479	2.474	
	Ebn at Noon	813	854	865	855	860	854	847	835	830	800	762	772	
	Edh at Noon	67	89	109	128	129	127	126	118	101	82	65	58	
All-Sky Solar Radiation	RadAvg	1.22	2.08	3.15	4.15	4.88	5.38	5.37	4.78	3.44	2.34	1.30	1.01	
	RadStd	0.14	0.27	0.37	0.45	0.53	0.40	0.45	0.54	0.44	0.23	0.19	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (110 neighbors)		+0.54	N/A	N/A	+0.87	+0.48	+0.39	-109	-150	+94	+57

Nomenclature: See separate page